Lecture 21 Scribe

L21_L22.pdf
PDF

Act as a professional Lecture Scribe for CSE 400: Fundamentals of Probability in Computing and generate a comprehensive, exam-ready lecture scribe for Lecture 21: Introduction to Tail Bounds and Markov's Inequality based strictly on the relevant sections of the provided context in L21_L22.pdf. In accordance with the mandatory Chain-of-Thought (CoT) instructions, you must reason step-by-step through the definition of a "tail" ($P\{X > x\}$), the motivation behind seeking tail bounds for system guarantees, and the formal derivation and proof of Markov's Inequality. Ensure these reasoning steps are explicitly included in the final document so a student can follow the logical flow and understand how the inequality provides an upper bound on probability using only the mean. The scribe must begin with the objective statement: "Objective: This scribe serves as an academic reference for exam preparation for Lecture 21". All content and logical derivations must be traceable only to the provided lecture materials, including the specific examples and visualizations of tail probabilities, without introducing outside intuition or higher-order bounds not covered in this specific session. Reason through the constraints of Markov's Inequality, such as the requirement for a non-negative random variable, and make the logical dependencies explicit. Ensure all mathematical expressions and the proof steps are rendered while maintaining a formal, academic, and disciplined tone throughout.

CSE 400: Fundamentals of Probability in Computing

Lecture 21 Scribe: Introduction to Tail Bounds and Markov's Inequality

Objective

This scribe serves as an academic reference for exam preparation for Lecture 21.

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1. Introduction to Tail Probabilities

1.1 What is a Tail?

Let X be a random variable.
A **tail probability** refers to the probability that X takes values **far away from typical/expected values**, specifically:

$$P\{X > x\}$$

This represents the **right tail** of the distribution.

Step-by-Step Understanding

1. A random variable X has a distribution over possible values.
2. Most values cluster around the mean $\mathbb{E}[X]$.
3. The **tail region** consists of values much larger than the mean.
4. The expression:

$$P\{X > x\}$$

measures how likely it is that X exceeds a threshold x.

Interpretation

- If x is large, then $P\{X > x\}$ should ideally be small.
- This probability quantifies **extreme events**.

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2. Motivation for Tail Bounds

2.1 Why Study Tail Bounds?

From the lecture context:

> Systems often need guarantees on **worst-case behavior**, not just average performance.

Step-by-Step Motivation

1. In computing systems (e.g., servers), average load is not sufficient.
2. We must understand:
   - "What is the probability that demand exceeds capacity?"
3. This is exactly a **tail probability problem**:

$$P\{X > x\}$$

Case Study Insight

- Example: Predicting **worst-case demand on a server**
- Even if average demand is low:
  - A high tail probability → risk of overload
- Hence, we need **upper bounds** on tail probabilities.

Goal of Tail Bounds

We want to find:

$$P\{X > x\} \leq (\text{some computable upper bound})$$

Using **limited information**, often just:

- Mean $\mathbb{E}[X]$

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3. Markov's Inequality

3.1 Statement of Markov's Inequality

Let X be a **non-negative random variable** (i.e., $X \geq 0$).

For any $a > 0$:

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

3.2 Key Constraints

Why must $X \geq 0$?

Step-by-step reasoning:

1. The proof relies on bounding contributions of values $X \geq a$.
2. If negative values exist:
   - They could cancel out positive contributions.
3. This breaks the inequality logic.
4. Therefore, **non-negativity is required**.

3.3 Intuition Behind the Inequality

We want to bound:

$$P(X \geq a)$$

Using only:

$$\mathbb{E}[X]$$

Key Idea

- If many values of X are large (≥ a), then:
  - The average $\mathbb{E}[X]$ must also be large.
- If the average is small:
  - The probability of large values must be small.

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4. Proof of Markov's Inequality

Step 1: Start with Expectation Definition

$$\mathbb{E}[X] = \sum_x x \cdot P(X = x)$$

Step 2: Split the Sum

Divide into two parts:

$$\mathbb{E}[X] = \sum_{x<a} xP(X = x) + \sum_{x \geq a} xP(X = x)$$

Step 3: Lower Bound the Second Term

For all $x \geq a$:

$$x \geq a$$

So:

$$xP(X = x) \geq a \cdot P(X = x)$$

Step 4: Apply This Bound

$$\sum_{x \geq a} xP(X = x) \geq \sum_{x \geq a} aP(X = x)$$

$$= a \sum_{x \geq a} P(X = x)$$

$$= a \cdot P(X \geq a)$$

Step 5: Combine Everything

Since expectation includes both parts:

$$\mathbb{E}[X] \geq a \cdot P(X \geq a)$$

Step 6: Rearranging

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

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5. Interpretation of the Result

What Does This Mean?

- The probability that X exceeds a is bounded by:

$$\frac{\mathbb{E}[X]}{a}$$

Key Insight

- Only the **mean** is needed.
- No need for:
  - Variance
  - Distribution shape

Behavior

- As a increases:

$$P(X \geq a) \downarrow$$

- Larger thresholds → smaller upper bounds

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6. Role in Tail Bounds

Markov's Inequality provides:

$$\text{A simple upper bound on } P\{X > x\}$$

Why Important?

1. Works with minimal information (only mean)
2. Provides **worst-case guarantees**
3. Foundation for:
   - Chebyshev's Inequality
   - Chernoff Bounds (introduced later)

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7. Limitations

Loose Bound

- The inequality can be **very loose**.
- It does not use:
  - Variance
  - Distribution details

But Still Useful

- When only $\mathbb{E}[X]$ is known
- In early-stage system analysis

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8. Summary

Key Concepts

- Tail probability:

$$P\{X > x\}$$

- Important for:
  - Worst-case system analysis
- Markov's Inequality:

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

Logical Flow Recap

1. Define tail probability
2. Motivate need via system guarantees
3. Use expectation to bound extreme events
4. Derive inequality step-by-step
5. Interpret and apply

Exam Tips

- Always mention **non-negativity condition**
- Write **full proof steps clearly**
- Explain **why expectation bounds probability**
- Connect to **tail behavior and system guarantees**

End of Lecture 21 Scribe.